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Melanotaenia susii, male in mating colouration.

Photo: Hans-Georg Evers

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A review of the genus *Kiunga* (Teleostei: Pseudomugilidae), with notes on distribution, potential environmental impacts and recommendations for research and conservation

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Abstract

Available information on the genus *Kiunga* is reviewed. *Kiunga* has two species, both of which occur in the Kiunga area, in the Fly River catchment, Western Province, Papua New Guinea. *Kiunga ballochi*, described in 1983, has recently been re-assessed as *Endangered*, and *K. bleheri*, described in 2004, as *Critically Endangered* (IUCN, in press). Although both species occur in several creeks in the vicinity of Kiunga, the spatial extent of their distribution and environmental variables that determine their distribution remain poorly known. Increased anthropogenic pressure in the Kiunga area, coupled with potential natural threats may endanger remaining populations of *Kiunga*. Recommendations for further research and approaches for conservation are made.

Introduction

New Guinea (NG), in the Indo-Pacific Region is an area that harbours a high number of endemic freshwater fish species (Polhemus *et al.*, 2004), with new endemic species documented regularly over the last two decades (e.g. Allen & Unmack 2008, Allen *et al.*, 2008a, Allen *et al.*, 2014, Allen *et al.*, 2016). The Fly River, in the Western Province of Papua New Guinea, is the largest river in NG, harbouring more than 108 species of freshwater fishes, and a higher number of endemic species than any other river system in NG (Allen *et al.*, 2008b). The Fly, together with rivers in the Trans-Fly Region share a number of species with northern Australia (Allen, 1991) including members of the family Pseudomugilidae or Blue-eyes.

Pseudomugilidae are among the smallest (< 5 cm fork-length) bony fishes found in freshwater and brackish estuaries of southern New Guinea and northern Australia (Allen *et al.*, 2008b). Nineteen species in three genera are currently known, of which 12 species in two genera (*Kiunga* and *Pseudomugil*) occur in New Guinea. Three species of *Pseudomugil* are shared with Australia, and three species of *Pseudomugil* and a monotypic genus (*Scaturiginichtys vermeilipinnis*) are endemic to Australia (Ivantsoff *et al.*, 1991, Fairfax *et al.*, 2007) (Table 1).

The genus *Kiunga*, first recorded from creeks north of the town of Kiunga, Western Province, Papua New Guinea (PNG) (Figure 1), is endemic to PNG and currently has two species; *K. ballochi* (Kiunga Blue-eye) and *K. bleheri* (Bleher's Blue-eye). The genus

was discovered in 1982 when *K. ballochi* was first recorded (Allen, 1983). Subsequently, a second species, *K. bleheri* was described (Allen, 2004). The genus differs from *Pseudomugil* based on a number of distinctive internal and external features including having vestigial and inconspicuous dorsal fins, a rigid keel-like structure mid-ventrally between pelvic and anal fins, and variations in skeletal features (Allen, 1983, Saeed *et al.*, 1989, Allen *et al.*, 2008b). The current knowledge of *Kiunga* shows it has a very limited distribution, and as such, may be susceptible to localised extinction. In this paper we summarise the current knowledge of the distribution of species in the genus *Kiunga*, and discuss potential threats and future conservation.

Kiunga ballochi Allen 1983 (Kiunga Blue-eye)

Kiunga ballochi (Plate 1) was first discovered during baseline surveys conducted for the Ok Tedi mine (Allen, 1983). Specimens were collected from “several small creeks...located approximately 30–50 km north of Kiunga...” (Allen, 1983). The creeks, accessed from a road being built to service the mine, were ‘pristine’, with no obvious disturbance apart from the new road crossings. Sampling was by seine net and took place during an El Niño-associated dry period when fish were concentrated and plentiful. Because the location of the type locality was estimated from a topographic map (Allen, 2007), it was difficult to determine its exact position, but subsequent surveys by various individuals and research organisations (Tables 2 & 3) failed to collect *K. ballochi* from creeks in the vicinity of the type locality, between Kiunga and Ningerum. Three trips were made to the area by Heiko Bleher (in 1991, 1994

Blue-eyes	Distribution
<i>K. ballochi</i> , <i>K. bleheri</i> <i>Pseudomugil</i> (<i>Popondichthys</i>) <i>connieae</i> , <i>P. (p.) furcatus</i> , <i>P. ivantsoffi</i> , <i>P. luminatus</i> , <i>P. majusculus</i> , <i>P. novaeguineae</i> , <i>P. paludicola</i> , <i>P. paskai</i> , <i>P. pellucidus</i> , <i>P. reticulatus</i>	NG endemic
<i>P. cyanodorsalis</i> , <i>P. signifier</i> , <i>P. mellis</i> <i>Scaturiginichtys vermeilipinnis</i>	Aust. endemic
<i>P. gertrudae</i> , <i>P. inconspicuus</i> , <i>P. tenellus</i>	NG & Aust.

Table 1. Regional distribution of Blue-eyes. [NG = New Guinea, Aust. = Australia]

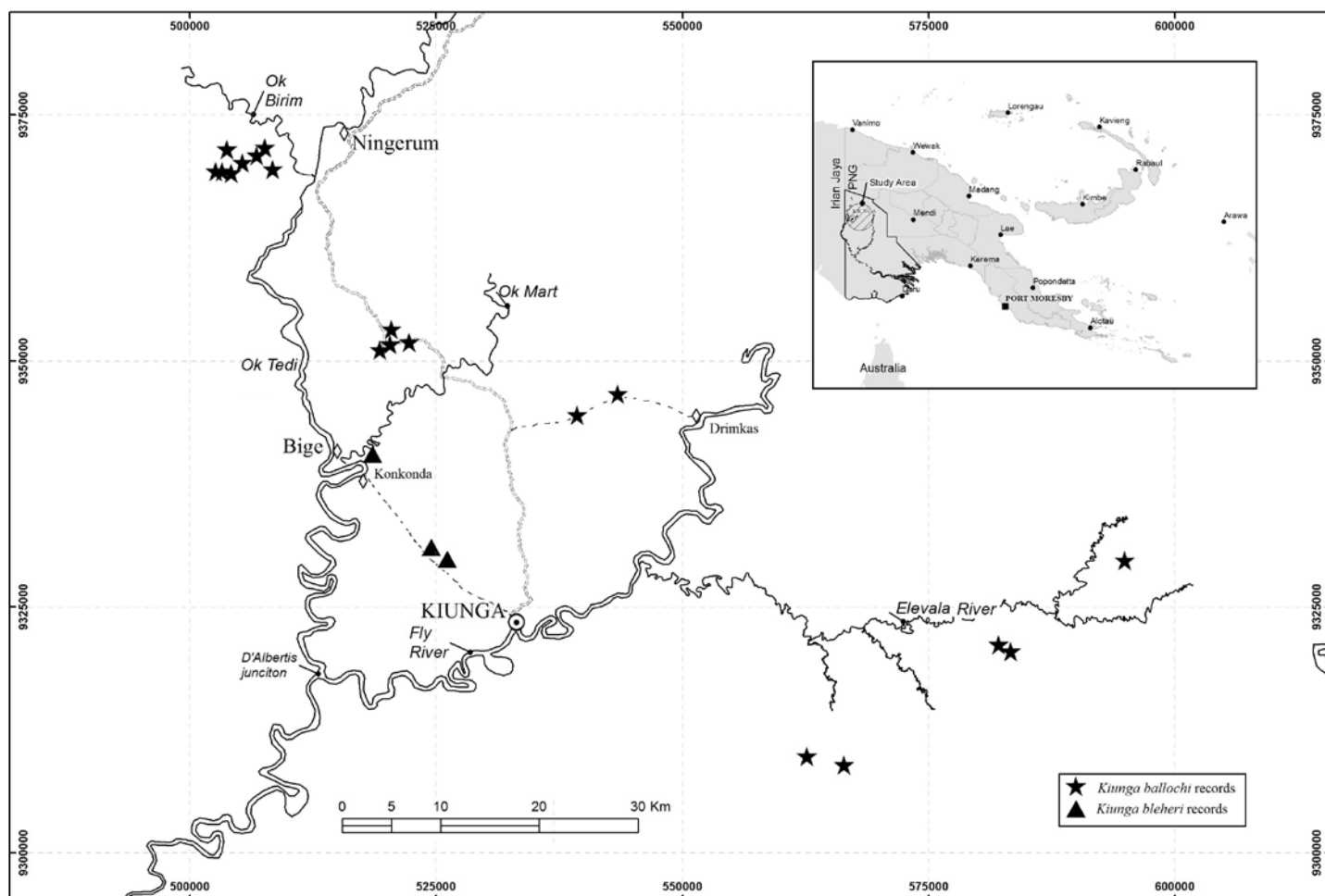


Figure 1. Locations around Kiunga-Ningerum, Western Province, Papua New Guinea, where *Kiunga ballochi* and *K. bleheri* have been collected since each was first discovered.

and 2003) in an attempt to secure specimens for the aquarium trade. He was unsuccessful in finding *K. ballochi*, despite rigorous sampling at the locality from where it was previously known and considered common. During the 1991 and 1994 survey however, specimens of a second species of *Kiunga* were collected (see below). The inability to relocate *K. ballochi* lead, in 2004, to the species being presumed extinct in the wild (see Allen, 2007).

In 2005, the junior authors (MY & AWS) undertook a fish diversity survey of the Fly River system, including 26 creeks along the Kiunga to Tabubil highway. Sampling was by seine net and electrofisher, but failed to record any specimens of *K. ballochi* or its congener. In 2007 *K. ballochi* was subsequently listed in the International Union for Conservation of Nature (IUCN) Red List as *Critically Endangered* because of the perceived severe reduction in number of fish (abundance) since its initial collection in 1982 (Jenkins *et al.*, 2009). It was not until 2007 that *K. ballochi* was ‘rediscovered’ (see Allen, 2007), during a survey of the Fly system to collect specimens and photographs for a new book on the freshwater fishes of the Fly River (Allen *et al.*, 2008b). During that survey, and after considerable effort sampling many creeks along the same highway, four specimens of *K. ballochi* were eventually caught in a fine mesh seine in a creek 42.5 km from Kiunga (Plate 2, see Allen, 2007). In 2011 the junior authors (MY & AWS), along with Gerry and Mark Allen conducted a follow-up fish diversity survey of the Fly system, in which 63 locations were sampled, including 21 creeks in the Kiunga-Tabubil area. This survey discovered a second population of *K. ballochi*, in a creek along a newly constructed road running from the Kiunga-

Tabubil Highway to the Bige dredge facility on the lower Ok Tedi (Figure 1, Plate 3).

Both new records of *K. ballochi* were from within a few km of the type locality along the Kiunga-Tabubil Highway (Allen, 1983). In 2014 the known distribution of *Kiunga* was expanded, when several specimens of what nominally appeared to be *K. ballochi* were discovered during baseline surveys by Horizon Oil (Ben Charles, Manager Environment & Safety, Horizon Oil, pers. comm.). Populations were recorded from three creeks to the east of Kiunga, in the Elevala River subcatchment (Plate 4), with an additional population from a tributary of the Upper Fly River, north of Kiunga. These specimens were identified as *Kiunga ballochi* by Gerry Allen and AWS from scanned photographs, but no specimens were available to confirm species identity. However, these were the first records of *Kiunga* to the east of the Fly River, and extends the known range of *Kiunga* by approximately 40 km eastwards, and beyond the Fly River main channel. The distribution was further extended in 2015 when *K. ballochi* was recorded from eight of 21 creeks sampled in the Ok Birim area to the west of the lower Ok Tedi by AWS and MY as part of baseline surveys of the area (Plates 5–7). These new populations are 12–15 km west-south-west of Ningerum, and two km from the border with Papua Province. They are also the first records of *K. ballochi* to the west of the Ok Tedi main channel, previously considered a potential barrier to their dispersion.

In October 2018, AWS & MY, with support from Ok Tedi Mining, revisited sites sampled by Horizon Oil in the Elevala



Plate 1. Adult male *Kiunga ballochi* with its distinctive yellow fins tipped off with dark margins for dorsal, caudal and anal fins.

Photo: Mark Allen

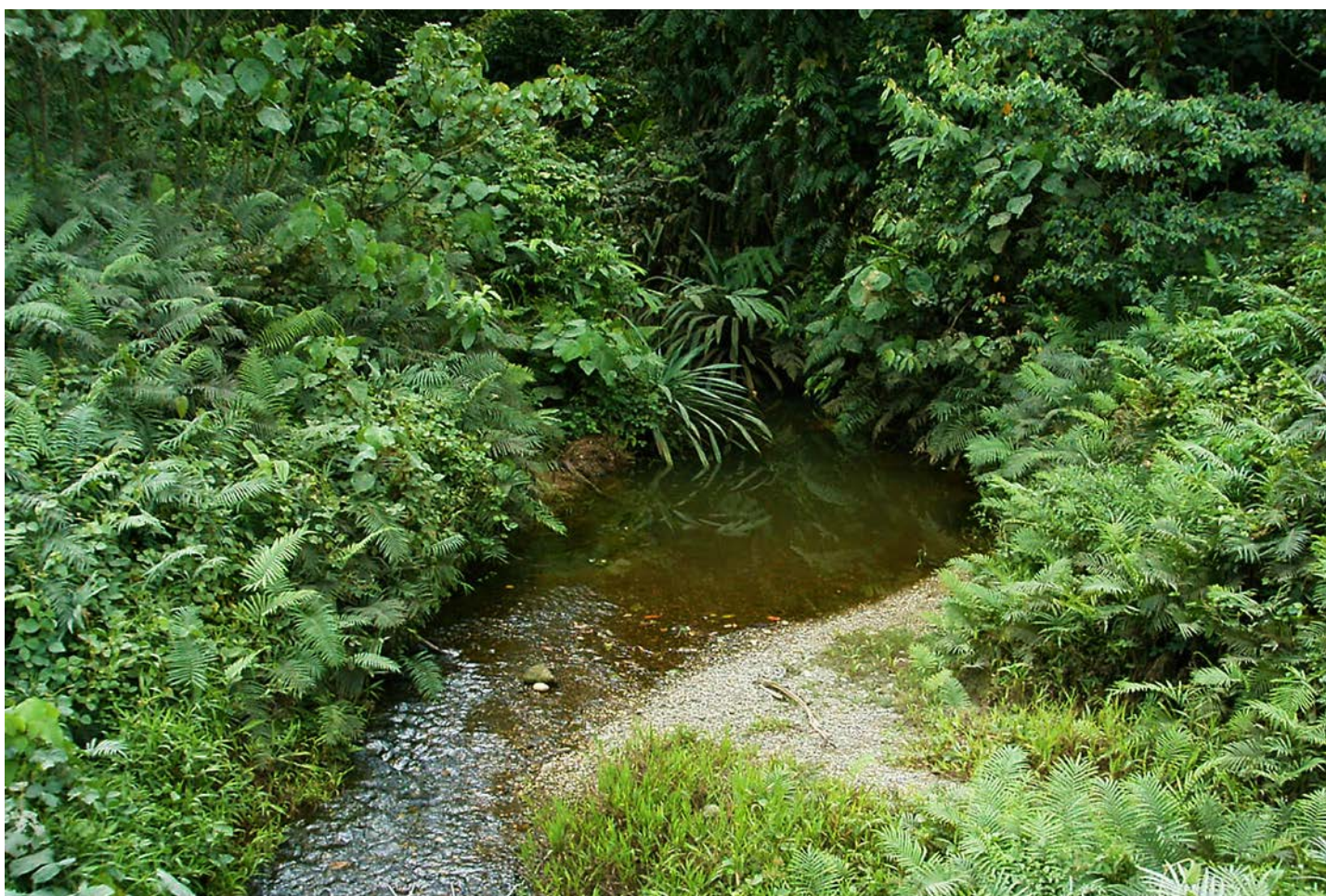


Plate 2. Creek at KM42.5 on Kiunga-Tabubil Highway where *Kiunga ballochi* was 'rediscovered' in 2007.

Photo: Andrew Storey



Plate 3. Collecting *Kiunga ballochi* from a creek on the Bige access road, with Gerry Allen in the foreground, and authors Pita Amick and Andrew Storey, with local assistants, in the background. Photo: Mark Allen



Plate 4. Taki Creek in the Elevala subcatchment; site where the eastern population of *Kiunga ballochi* was recently collected (October 2018). Photo: Andrew Storey



Plate 5. Unnamed tributary in Ok Birim catchment where *Kiunga ballochi* was collected.

Photo: Andrew Storey

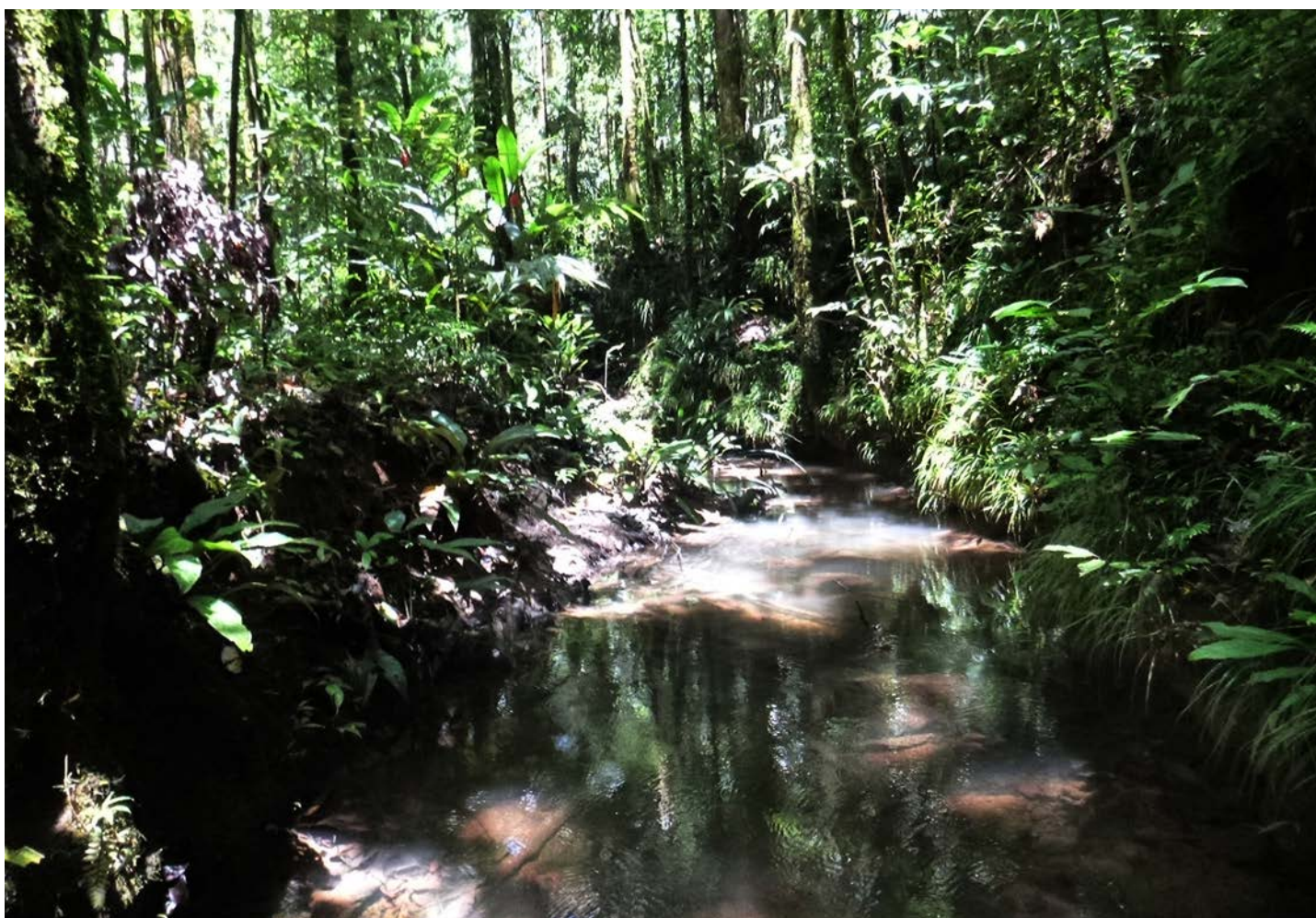


Plate 6. An unnamed tributary of the Ok Birim, on the west bank of the Ok Tedi, where *Kiunga ballochi* was collected in March 2015.

Photo: Andrew Storey



Plate 7. An unnamed tributary in Ok Birim catchment where *Kiunga ballochi* was collected.

Photo: Andrew Storey

subcatchment and collected specimens of *Kiunga* from one site, Taki Creek, as well as from two creeks along the newly constructed Drimkas Highway, a new road from Kiunga to the upper Fly/ Palmer River confluence at Drimkas. The specimens from Taki Creek (Plate 8) looked sufficiently different to *K. ballochi*, based

on colouration and size to potentially be a different species (Gerry Allen, pers. com.), and await genetic sequencing and morphological examination. The discoveries in 2014, 2015 and 2018 significantly increased the distribution of *K. ballochi*, and it seems likely a.) populations will occur in undisturbed creeks



Plate 8. The eastern form of *Kiunga ballochi*, collected from Taki Creek in the Elevala subcatchment, sampled October 2018.

Photo: Andrew Storey

Year of survey	Lead scientists and purpose
1982	Gerry Allen, with John Paska, sampled creeks along the new Kiunga-Tabubil highway as part of baseline surveys during construction of the Ok Tedi mine.
1991	Heiko Bleher, creeks along the Kiunga-Tabubil and Kiunga-Konkonda highways in an attempt to capture specimens of <i>K. ballochi</i> for aquarium trade
1994	Heiko Bleher, creeks along the Kiunga-Tabubil and Kiunga-Konkonda highways in an attempt to capture specimens of <i>K. ballochi</i> for aquarium trade.
2003	Heiko Bleher, a third visit to creeks along the Kiunga-Tabubil and Kiunga-Konkonda highways to attempt to capture specimens of <i>K. ballochi</i> but also gather new specimens of <i>K. bleheri</i> .
2005	Andrew Storey, Ross Smith and staff of Ok Tedi Mining Limited, to conduct fish diversity survey of the Fly River, sampling 50 sites throughout the system, capturing 84 native and 3 exotic species.
2007	Andrew Storey, Gerry and Mark Allen and Markson Yarrao to collect specimens and gather photographs for a proposed new book on the freshwater fishes of the Fly River
2011	Andrew Storey, Gerry and Mark Allen and Markson Yarrao to conduct fish diversity survey of the Fly River, sampling 57 sites throughout the system, capturing 84 native and 5 exotic species.
2014	Baseline surveys of creeks north of Kiunga and in the Elevala River area, east of Kiunga by Horizon Oil.
2015	Andrew Storey and Markson Yarrao, 25 sites in the Ok Birim area as part of a baseline survey, recording 34 species of fish.
2018	Andrew Storey & Graham Mesa, 3 sites sampled by Horizon Oil in the Elevala subcatchment, and 6 creeks along the Drimkas Highway from Kiunga to the upper Fly

Table 2. Summary of past surveys for Kiunga in the Kiunga-Ningerum area.

between these currently known eastward and westward limits of its distribution and b.) the distribution will extend westwards into Papua Province, Indonesia, and further eastwards towards the Strickland River and even beyond. Based on these new findings, a recent reassessment of the conservation status of New Guinea fishes, re-assessed *K. ballochi* as *Endangered* (IUCN, in press).

Kiunga bleheri Allen 2004 (Bleher's Blue-eye)

Kiunga bleheri (Plate 9) was reportedly first collected by Heiko Bleher in 1991 from Tare Creek, about 10 kilometres north-west of Kiunga, close to Konkonda on the Konkonda Road, Western Province, PNG (Figure 1, Plate 10) (Tappin 2015), when it was mistakenly identified as *K. ballochi*, but the live specimens were lost. In 1994 they were collected again, by Heiko Bleher, from a creek 26 km north-west of Kiunga, but the two individuals didn't breed (Allen, 2004). In 2003 Heiko Bleher managed to catch a larger number, again from Tare Creek, and these were successfully bred and established in the aquarium trade, and subsequently described by Allen (2004). The 2005 fish diversity survey (AW Storey, unpub. dat.) sampled three creeks along the Konkonda Highway, including the type locality Tare Creek, but failed to collect *K. bleheri*. The species however was captured during the 2007 survey (Allen *et al.*, 2008b) and again during the 2011 diversity survey, but on both occasions only from the type locality sampled by Bleher in 1994 and 2003 (Allen, 2007). Despite repeated surveys, *Kiunga bleheri*, is still only known from a few small creeks in the Kiunga area. On this evidence,

K. bleheri appears extremely restricted in distribution compared to *K. ballochi*, with only two known populations recorded in close proximity, from a comparable sampling effort (Tables 2 & 3). Based on these findings, the recent reassessment of the conservation status of New Guinea fishes, assessed *K. bleheri* as *Critically Endangered* (IUCN, in press).

Habitat conditions

Based on their current known distributions, and conditions at the locations from which *K. ballochi* and *K. bleheri* have been collected, the two species are allopatric and probably fill similar ecological niches in terms of food preference, habitat use and their tolerance to variation in water quality. All collecting sites are small and shallow (5 m wide x 1 m deep), slow-flowing creeks at elevations around 80 ($\pm$ 20) m asl, with a daytime temperature range of 24–27°C, pH near neutral to slightly acidic (pH 5.7–7.8), and very fresh (EC < 30 μ S/cm) (Table 4). During field work in 2011 (A.W. Storey unpubl. dat.), *K. ballochi* was collected from the riffle-pool interface of a slow-flowing creek along the road to Bige; *K. bleheri* were collected in a forest-fringed pool along the Konkonda Road. The most extensive collection of *K. ballochi*, in 2015, and again the collections in 2018 were all from clear to slightly turbid, slow-flowing creeks with 100% canopy cover, dense riparian and trailing vegetation, substantial large woody debris loads, predominantly organic substrate, with moderate presence of submerged macrophyte beds (Plates 4–7).

Field work	1982	1991	1994	2003	2005	2007	2011	2014	2015	2018
<i>K. ballochi</i>	√	x	x	x	x	√	√	√	√	√
<i>K. bleheri</i>	x	√	√	√	x	√	√	x	x	x

Table 3. Summary of recordings of *K. ballochi* and *K. bleheri* by past surveys of creeks in the Kiunga area, Western Province, Papua New Guinea. [Symbols: √ = recorded, x = not recorded]

Potential environmental impacts

Given the limited spatial distribution and apparently small population sizes of both species at locations where they were found, and increasing anthropogenic disturbance across much of their distribution, combined with the threat from climatic change, both species, but *K. bleheri* in particular, are likely at risk of extinction. The following sections outline some of the potential threats to the genus based on the authors observations of developments in the area.

Habitat degradation

When first sampled in 1982, *K. ballochi* were “...found in large aggregations and easily netted” (Allen, 2007), and the creeks being sampled were near pristine, with the only disturbance being the new road crossings for the road from Kiunga to Tabubil. Although the construction of the road initially allowed access to previously remote and inaccessible creeks, now the road, combined with the employment and compensations opportunities provided by the Ok Tedi mine, has attracted an ever-increasing human population to the area. The ‘near pristine’ creeks of 1982 are a thing of the past, especially along the highway, with small communities now situated on most creek crossings, with the creeks used for water supply, washing and disposal of ‘waste’. Often when sampling these creeks in 2005, 2007 and 2011, we were watched by interested villagers, who had been washing or fishing in the creeks. The increasing population also brings clearing for gardens and rubber plantations, all of which put pressure on the waterways. Undoubtedly there are ‘near pristine’ creeks deep in the rainforest, away from the highway, and perhaps the difficulty of access into these areas that prevents us from sampling, will also protect populations of *Kiunga*

that most likely still persist. The increased anthropogenic pressure at the road crossings would appear to have adversely affected the previous ‘large aggregations’ found at these locations, suggesting this genus is quite sensitive to such disturbance. It is interesting to note that the new populations of *K. ballochi* found in 2011 in a creek on the Bige access road, and again in 2018 in creeks on the new Drimkas road, were along relatively new roads and the creeks were still undisturbed when sampled. It will be interesting to see how these populations fair over time. Similarly, the eight sites in the Ok Birim area which held *K. ballochi* in 2015 are very remote, accessed by helicopter and foot, and being situated on the west side of the Ok Tedi they are currently isolated from the highway and associated developments. One would hope these populations are reasonably secure.

Increased total suspended solids in waterways

The initial construction of the Kiunga-Tabubil road, as well as local roads such as the Kiunga-Konkonda, and Rumginae and Bige access roads would all have initially produced high sediment loads into the various creeks they cross. The everyday use of these unpaved, gravel roads, as well as activities such as clearing for plantations and gardens would also produce run-off with high sediment levels during the frequent rainfall events. Although the sensitivity of *Kiunga* to sediment loads is not known, they are likely sensitive to elevated TSS given the normally clear nature of the lowland, low gradient rainforest creeks they inhabit. It is likely elevated sediment would smother fish eggs laid attached to submerged vegetation or even smother the spawning habitat itself, resulting in population decline.



Plate 9. Adult male *Kiunga bleheri* showing almost transparent fins, with black tips for dorsal and anal fins.

Photo: Gerald Allen



Plate 10. Tare Creek, type locality of *Kiunga bleheri* on Kiunga-Konkonda Highway, being sampled in August 2005.

Photo: Andrew Storey

Water quality coupled with climate change and extreme weather conditions

Change in local weather patterns due to global warming may have effects on the general water physico-chemistry (e.g. pH, temperature, conductivity, dissolved oxygen). Although change in water quality is occurring over time as a result of natural variability due to wet and dry periods, a steady increase or decrease in certain parameters due to climate change may negatively affect *Kiunga* if they do not adapt accordingly. Due to their relatively smaller size, *Kiunga* might not physiologically adapt to rapid changes in their surroundings. Extreme weather condition such as the El Niño droughts (usually prolonged dry weather), which are predicted to increase in frequency and severity, may severely affect *Kiunga* and potentially lead to localised extinctions if watercourses dry up.

Unexpected environmental incidents by mining, oil and gas companies

The development of mining, oil and gas in the region also raises the potential for direct impact from these activities on streams supporting populations of *Kiunga*. The road between Kiunga and Tabubil is the conduit for the copper concentrate pipeline from the Ok Tedi mine to the port facility at Kiunga. It is also the conduit for the waste pyrite concentrate pipeline from the mine to the burial area at Bige. Oil and gas activities in the region by several companies, including exploration programs, as well as production activities also pose a threat. All these activities use convoys to transport goods such as fuel, oil, process chemicals and explosives (ammonium nitrate) along the highways. Ruptures or sabotage of the pipelines, or traffic accidents along the highway could result in spills into adjacent streams, potentially having negative impacts on populations of *Kiunga* due to habitat degradation, or reduced water quality (i.e. spills of hydrocarbons, copper or pyrite concentrate, explosives or any other harmful chemicals).

Edge and barrier effects

Degradation of habitats and water-use activities may have edge-effects on the occurrence of *Kiunga*. Although populations of *K. ballochi* occur to the east of the Fly River and to the west of the Ok Tedi, the only known population of *K. bleheri* is 'land-locked' between these systems, but is also in the area where most impacts from development are occurring. It is likely the Ok Tedi currently forms a barrier to future dispersal, given the high sediment loads and strong current (Smith & Morris 1992). The main channel of the Fly River may also form a barrier to eastward dispersion given its size. These barriers may prevent *K. bleheri* dispersing and thereby avoiding pressures from changes in land use and water quality.

Impacts from introduced fishes

From having no introduced fish species in the early 1990s, the Fly River system now supports five exotic species (Storey *et al.*, 2002, Allen *et al.*, 2008b). *Channa striata* (Striped Snakehead) is a voracious piscivore, and occurs throughout the system below ~ 100 m elevation. The observed behaviour of *Kiunga ballochi* of swimming in small schools in sun-lit areas amongst beds of macrophyte, in slow flowing creeks (AW Storey, pers obs), would make them an easy target for *C. striata*. *Channa striata* were regularly caught from the small creeks along the Kiunga – Tabubil highway during fish diversity surveys (AW Storey, unpub. dat.). *Oreochromis niloticus* (Nile Tilapia) and *Anabas testudineus* (Climbing Perch) also occur in lowland creeks within the known range of the species of *Kiunga*. *Oreochromis niloticus* are potentially piscivorous when food is scarce, and *A. testudineus* are, to various degrees, competitors for food and habitat (Storey *et al.*, 2002). Both species feed on detritus and macrophytes, and will likely accidentally, or intentionally ingest eggs if accessible. The remaining introductions, including, *Cyprinus carpio* (Common Carp) and *Clarias batrachus* (Walking Catfish) are not yet common in the lowland reaches, and were seldom recorded in the area, but both would be competitors for space and food should numbers and distribution increase.

Natural fish disease

Fish, like any other living organism, may contract disease, which may endanger populations and species. Native fishes may also contract disease from introduced fish species. Clinical study is lacking on fishes occurring in the wild (Li *et al.*, 2002, Noga 2011) as more emphasis is on disease associated with aquaria and farmed fish (e.g. Noga 2011). Of the five introduced fish species in the Fly River system, only *C. carpio* and *O. niloticus* have been introduced officially, albeit to aquaculture ventures from which they have escaped into the wild. *Channa striata*, *A. testudineus* and *C. batrachus* have all come across the border from Papua Province as a result of introductions by trans-migrants to that part of New Guinea. This means that the stock introduced into the system have not been properly quarantined *per se* (Kolkolo, 2005) and therefore may carry pathogens that may be harmful to native species, including *Kiunga*. There is no record of fish disease in the Fly River system at the moment, however, it has not been studied.

RECOMMENDED FUTURE STUDIES AND CONSERVATION ACTIONS

Document the spatial distribution of *Kiunga*

Although there are several publications specifically dealing with *Kiunga* (e.g. Allen, 1983, Allen, 1991, Crockford, 1997, Allen, 2004, Allen, 2007, Tappin, 2011, 2015), they all still lack clear information on spatial distribution of both species. The majority of our current knowledge of this genus is garnered from baseline surveys and impact assessments for mineral, oil, gas and construction companies, which are often held as internal reports and are not publicly available. The current paper assimilates all known records of both species of *Kiunga*, and greatly expands our knowledge of the distribution of both species. Creeks in a similar geographical (geomorphology and natural environment) setting between the Strickland and Digul rivers, at least within the range of 100 ± 30 m a.s.l. may well support populations of the two currently known species of *Kiunga*, or even reveal additional unknown species. But the general inaccessibility of most creeks within these areas will limit our ability to survey these areas, but that remoteness in itself may protect them. Evidence from the *Kiunga*-Tabubil road suggests this species is sensitive to anthropogenic disturbance. Of concern, however, is extensive clearing, development of road networks, plantations and population growth in Papua Province close to the border with PNG. The new records of *K. ballochi* to the west of the Ok Tedi, and close to the Indonesian border suggest the species may extend into Papua Province. The development in this area, likely with little environmental regulation, could mean unknown populations are lost before they have been found!

Long term data on physio-chemistry and population trends

General information on the ecology of *Kiunga* is lacking. Having a monitoring program that includes population studies and

monitoring of physico-chemical variables of creeks could show trends and provide important community information, but would certainly expand our knowledge of the habitat requirements of the genus. Vital, baseline information on basic water quality parameters recorded by Allen (1983) and Allen (2004), and Storey (unpub. dat.) for creeks where *Kiunga* have been collected is provided (Table 4). A monitoring program has been implemented for the endangered *Scaturiginichtys vermeilipinnis* (Red-finned Blue-eye) in northern Australia (Fairfax *et al.*, 2007) and a similar program could be implemented for both *Kiunga* species.

Genetic analysis of current populations

Specimens of *K. ballochi* from the Ok Birim survey were compared genetically to specimens of *K. ballochi* collected from creeks to the east of the Ok Tedi, as well as the congener *K. bleheri*. The Ok Birim specimens were found to be genetically distinct to the known *K. ballochi* specimens (4% divergent), but were closer to *K. ballochi* than *K. ballochi* and *K. bleheri* were to each other (6% divergent), and most different to *K. bleheri* (8%). The Ok Birim *Kiunga* specimens were considered sufficiently distinct from the known *K. ballochi* specimens as to perhaps constitute a new species. However, further analyses including morphometric studies would be required to confirm that the specimens collected from the Ok Birim creeks do in fact represent a species new to science (Dr Peter Unmack, University of Canberra, pers. comm.). The divergence is not unexpected given the geographic separation, with the Ok Tedi main channel likely providing an impediment to east-west migration and associated gene flow. However, within the eight populations sampled from the Ok Birim there was minimal genetic difference, indicating these population are well mixed.

As noted above, *K. ballochi* from the original 2014 Elevala survey were informally identified from photographs. Although similar looking to *K. ballochi* from known populations, the colouration was different and they could well be a different species (Gerry Allen, pers. comm.). The new specimens sourced from the Elevala in 2018 support this contention (Plate 8) and await genetic and morphometric analysis to confirm their status.

Impacts of introduced fish on *Kiunga*

No study has been conducted on the impacts of the introduced fishes in the Fly River system, except one investigation on the Climbing Perch (Storey *et al.*, 2002). A specific study focusing on interactions of introduced fish species on *Kiunga* could provide information on the extent of impacts introduced fish have on *Kiunga*, and also on other native fish species.

Breeding programs

Atheriniformes (Blue-eyes and relatives) are good aquarium fishes. New Guinea Blue-eyes have been collected by aquarists,

	Temp. (°C)	pH	Conduct. (µS/cm)	Habitat	Elev (m)	Year rec.
<i>K. ballochi</i>	24.5	7.8	–	clear, shallow rainforest streams	~100	1982
<i>K. bleheri</i>	27	5.9	29	clear, shallow rainforest streams	~100	2003
<i>K. ballochi</i>	24 – 27	5.6–7.1	4–19	Clear, slow-flowing, foothill creeks with 100% canopy cover and moderate submerged macrophyte	70–100	2015

Table 4. Summary of recordings for physico-chemical parameters and brief description of riparian vegetation as provided by Allen (1983, 2004) and Storey (unpub. dat.).

taken overseas, and bred in captivity for commercial purposes. This includes both species of *Kiunga* in the past. No attention has been given to the remaining native stock, or to establishing breeding populations from new wild populations to maintain genetic diversity. As infrastructural developments in the Kiunga – Ningerum areas are increasing, and with the ever-increasing human population and associated local, land and water use activities, coupled with the potential natural threats, a serious decline, even extinction of the genus *Kiunga* is possible, and *K. bleheri* in particular given its limited distribution. Although National Fisheries Authority (NFA) of PNG focuses on commercial fisheries, an aquarium fishery also should be considered, not only for profit, but as a means for conservation of rare freshwater fish. Indeed, the Ok Tedi Development Foundation (OTDF) could consider this as a viable project in which to invest.

Conclusions

The Blue-eye genus *Kiunga* has only two representative species, *K. ballochi* and *K. bleheri*, both of which are endemic to the Kiunga area, PNG. *K. ballochi* was initially ranked *Critically Endangered* by IUCN in 2007 (Jenkins *et al.*, 2009), although more recent data suggests it is more widespread than previously thought, and its original ranking has been re-assessed as *Endangered*. Available information for *K. bleheri*, however, suggests this species has a very limited distribution, currently known from two creeks, one being the type locality. On this basis, and given the surrounding anthropogenic pressures, *K. bleheri* has been assessed as *Critically Endangered* under the IUCN Red List. The current paper expands our understanding of the current distribution of both species, but also indicates both are likely sensitive to a variety of both anthropogenic and natural environmental impacts. Although the studies listed here expand our knowledge of the distribution of both species, additional survey effort is required to fully understand their greater spatial distribution. Long term monitoring of *Kiunga* populations and monitoring of water quality parameters in their habitats may provide important data to further our understanding of *Kiunga* and their response to temporal variation in water physico-chemistry. Specific studies on impacts of introduced fishes on *Kiunga* should also shed light on the impacts of introduced fishes. All in all, increasing human activities in the Kiunga-Ningerum areas, coupled with potential natural threats, may affect the remaining *Kiunga* populations resulting in serious decline, or even extinction. An aquarium-based breeding program could help conserve *Kiunga*.

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